SIMONE ROMITI

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I am a computational physicist and ML developer with 8+ years of experience in building high-performance simulations, statistical inference analyses, and data science. My core expertise is in Monte Carlo methods, Bayesian statistics, deep learning, and HPC pipelines on multi-terabyte datasets. I have a proven track record of open-source software with CI/CD, automated testing, and versioned releases. I am comfortable translating technical complexity for diverse audiences, and experienced in leading projects with international teams.

Resident in Switzerland with B permit.

TECHNICAL SKILLS

Languages - C/C++, Python, R, Bash/GNU Linux

ML & Statistics - PyTorch, Physics-Informed Neural Networks (PINNs), VAEs, Diffusion Models, Bayesian Inference, Monte Carlo, Nested Sampling

Data & Scientific Python - NumPy, SciPy, Pandas, Matplotlib, Plotly, Streamlit, SymPy, Jupyter

HPC & Systems - CUDA, MPI, OpenMP, SLURM, EasyBuild

DevOps & Tools - Git, GitHub Actions, CI/CD, Docker, LaTeX, Quarto

Spoken Languages - Italian (native), English (proficient), German (basic)

WORK EXPERIENCE

University of Bern — Apr 2024 – Present
Postdoctoral Researcher — Bern, Switzerland

- **Reduced algorithm complexity from $O(N^6)$ to $O(N \log N)$** for a class of high-dimensional integrals via parallelized Fast Fourier Transforms
- **Bayesian inference pipeline** over multi-terabyte datasets: uncertainty quantification, systematic error budget, and model averaging; published with open-source implementation.
- **Physics-Informed Neural Networks (PINNs)**: developed novel application for the solution of the Schrödinger equation with periodic potentials; results published in peer-reviewed journal.
- **Application of Nested Sampling algorithm** for multi-modal density estimation and study of phase-transitions.
- **Principal Investigator** of two $O(10^6)$ GPU computing time allocations (CSCS ALPS, 2025 and GCS JUPITER,2026)

University of Bonn — Nov 2021 – Mar 2024
Postdoctoral Researcher — Bonn, Germany

- **Designed and operated large-scale Monte Carlo simulations** on international HPC clusters: tuning, statistical validation, and quality control of multi-terabyte datasets.
- **Developed a novel numerical method** achieving machine-precision accuracy for a class of constrained optimization problems; published in a peer-reviewed journal.
- **Hybrid Monte Carlo + Quantum Computing** method for the estimation of high-dimensional integrals.

EDUCATION

PhD in Physics | title awarded in April 2022 | — *Roma Tre University* | Nov. 2018 – Oct. 2021

M.S. in Physics | Grade: 110/110 *cum laude* | GPA: 29.85/30 | — *Roma Tre University* | Oct. 2016 – Oct. 2018

B.S. in Physics | Grade: 110/110 *cum laude* | GPA: 28.84/30 | Scholarship — *Roma Tre University* | Oct. 2013 – Jul. 2016

LEADERSHIP & ACHIEVEMENTS

Supervision of Masters and PhD students

Invited Speaker – Technical seminars at CERN and ECT* — 2025 – 2026

Workshop Organizer – International HPC & Quantum Computing workshop — ECT* | 2025